

A Deterministic Continuum Semiconductor Model for Telecom Laser Simulation

Version 2: Self-Consistent Carrier, Wave-Optics, and Thermal Manual

Abstract

This manual specifies Version 2 of the deterministic continuum model for telecom diode laser simulation. The model integrates electrostatic potential, carrier transport, lattice temperature, and electromagnetic wave propagation. In this version, the simplified scalar photon diffusion solver is replaced by a wave-optics formulation. EMSolver solves the 2D transverse vector Helmholtz equation, incorporating the analytical 1D longitudinal mode mirror losses. The local photon population is computed directly from the electromagnetic field intensity, closing the carrier stimulated recombination rate and lattice heat sources self-consistently.

1. Domain, Variables, and Material Masks

Let D be the simulated 2D waveguide cross-section of the telecom laser. The domain is decomposed into contact, cladding, MQW active layer, passivation, and metal regions. The active quantum wells are localized by a spatial mask $F_{\text{QW}}(x)$:

- $F_{\text{QW}}(x) = 1$ inside the MQW active region
- $F_{\text{QW}}(x) = 0$ elsewhere

Key variables in the system include:

- ψ : Electrostatic potential (V)
- n, p : Electron and hole densities (cm^{-3})
- E_{tr} : Transition energy for carrier-photon bookkeeping (eV)
- T : Lattice temperature (K)
- E_{Wave} : 2D complex electric field phasor (V/m)
- PhotonPopulation: Local photon population density (cm^{-3})

2. Electrostatic Potential: Poisson Equation

The electrostatic potential is governed by Poisson's equation, which determines the electric field driving carrier drift currents:

$$-\text{div}(\epsilon \cdot \text{grad}(\psi)) = q \cdot (p - n + N_{\text{D}}^{+} - N_{\text{A}}^{-})$$

$$E_0 = -\text{grad}(\psi)$$

Where q is the elementary charge, ϵ_0 is vacuum permittivity, and ϵ_r is the relative permittivity defined per material region and temperature.

3. Carrier Conservation: Drift-Diffusion Equations

Carrier densities are conserved via drift-diffusion continuity equations:

$$\frac{dn}{dt} = (1/q) \cdot \text{div}(J_n) - R$$

$$dp/dt = -(1/q) * \text{div}(J_p) - R$$

The conventional current densities are:

$$J_n = q * \mu_n * n * E_0 + q * D_n * \text{grad}(n)$$

$$J_p = q * \mu_p * p * E_0 - q * D_p * \text{grad}(p)$$

Where μ_n , μ_p are electron/hole mobilities, and D_n , D_p are the diffusion coefficients satisfying the Einstein relation: $D = \mu * k_B * T / q$.

4. Recombination Model

The common carrier recombination sink R is decomposed into four channels:

$$R = R_{sp} + R_{stim} + R_{SRH} + R_{Auger}$$

1. Spontaneous Radiative Recombination:

$$R_{sp} = F_{QW} * B_{sp} * [n * p - n_i^2] +$$

2. Stimulated Radiative Recombination:

$$R_{stim} = g(n, p, T) * \text{PhotonPopulation}$$

3. Shockley-Read-Hall Recombination:

$$R_{SRH} = [n * p - n_i^2] / [\tau_p * (n + n_1) + \tau_n * (p + p_1)]$$

4. Auger Recombination:

$$R_{Auger} = (C_n * n + C_p * p) * [n * p - n_i^2] +$$

5. Transverse Wave-Optics Model (EMSolver)

Instead of a simplified photon diffusion solver, the electromagnetic field phasor is solved directly using the frequency-domain vector Helmholtz wave equation in the 2D cross-section:

$$\text{curl}(\mu^{-1} * \text{curl}(E)) - \omega^2 * \epsilon * E + i * \omega * \sigma * E = i * \omega * J$$

Coupling features:

1. Carrier-dependent optical gain:

$$\sigma_{gain}(x,y) = C_{gain} * g(n,p,T) = C_{gain} * [n * p - n_i^2] / [n + p + 2 * n_i]$$

2. Spontaneous emission excitation source:

$$J_{sp}(x,y) = C_{sp} * \sqrt{R_{sp}} * e_{polarization}$$

3. Analytical longitudinal mirror loss:

$$\sigma_{mirror} = n_{index} * c_0 * \epsilon_0 * [1 / (2 * L_{cav}) * \ln(1 / (R_1 * R_2))]$$

4. Total effective conductivity:

$$\sigma = \sigma_{passive} + \sigma_{mirror} - \sigma_{gain}$$

5. Photon Population Export:

$$\text{PhotonPopulation}(x,y) = C_{scale} * |E(x,y)|^2$$

6. Lattice Heat Equation

The thermal equation conserves lattice energy, driven by electrical, recombination, and optical heat sources:

$$\rho * C_p * dT/dt = \text{div}(k_T * \text{grad}(T)) + Q_{gen}$$

$$Q_{gen} = Q_{Joule} + Q_{SRH} + Q_{Auger} + Q_{abs}$$

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Where:

- $Q_{\text{Joule}} = (J_n + J_p) \cdot E_0$ is Joule heating.
- $Q_{\text{SRH}} = E_{\text{tr}} \cdot R_{\text{SRH}}$ and $Q_{\text{Auger}} = E_{\text{tr}} \cdot R_{\text{Auger}}$ are recombination heating.
- $Q_{\text{abs}} = E_{\text{tr}} \cdot v_g \cdot \alpha_{\text{abs}} \cdot \text{PhotonPopulation}$ represents optical absorption heating.

Boundary conditions model package convective heat sink cooling at the electrical contacts: $-k_T \cdot \text{grad}(T)$.
 $n = h_c \cdot (T - T_{\text{sink}})$.